

Department of Mining Engineering
Indian Institute of Technology (ISM), Dhanbad

Examination: **END SEMESTER**

B. Tech (Open Elective) 6th Semester

Session: **2024-25**

Subject: **MNO 401 ROCK ENGINEERING**

Semester: **Winter**

Instructions: 1. Answer **ALL** questions

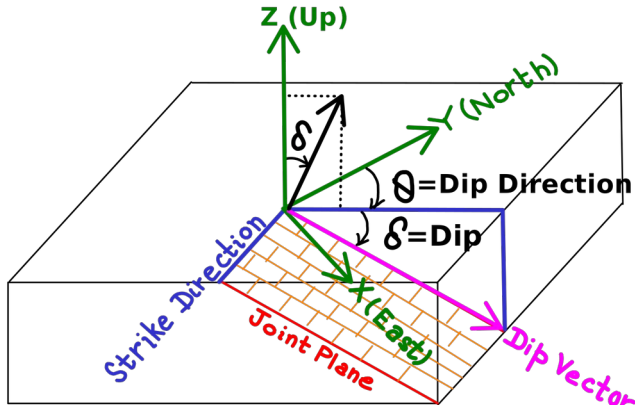
2. Neat & accurate sketches assists you in solving the problems logically and faster

3. Assume logical data, where necessary

4. Use **Rock Mechanics Sign Convention** to solve the problems

Time: **2 HOURS**

Max. Marks: **50**

Q. No.	Question		Marks
1	The orientation of a joint plane can be expressed in compass co-ordinates by dip amount (δ) and dip direction (θ). The vector normal to the joint plane can be converted to Cartesian coordinates (x,y,z). Where, - x-axis points towards the East direction - y-axis points towards the North direction - z-axis points towards the Up direction The relevant picture shows the relationships on the right. A joint plane is characterized by a dip amount of 30° and dip direction of 060° ($N60^\circ E$).	 The state of stress is given by the matrix below in xyz-coordinate system. Determine the magnitude of Normal traction on the joint plane. $\begin{bmatrix} 5 & 2 & 3 \\ 2 & 12 & 4 \\ 3 & 4 & 7 \end{bmatrix} MPa$	10
2	(a)	In case of infinitesimal strains in two dimensions, define the normal and shear strains and write the strain matrix in differential form.	2
	(b)	Prove that the strain matrix is a second order tensor	5
	(c)	The displacement field of a body is given by $\vec{U} = u(x, y, z) \vec{i} + v(x, y, z) \vec{j} + w(x, y, z) \vec{k}$ $= (9x^3 + 4) \vec{i} + (15y + 3x^2) \vec{j} + (6y^2 + 2z^2) \vec{k}$ Determine the deformed position of a point originally at (2, 5, -4)	3
3	(a)	Discuss the Coulomb failure criteria using a neat sketch of Mohr's circle. Further prove that the failure criterion in terms of the principal stresses can be expressed as $\sigma_1 = \frac{2c \cos \phi}{1 - \sin \phi} + \sigma_3 \frac{1 + \sin \phi}{1 - \sin \phi}$	5
	(b)	The state of stress in the xy-coordinate system is given by the following matrix $\begin{bmatrix} 20 & 0 \\ 0 & 5 \end{bmatrix}$ Using Mohr's circle construction, determine the normal stress and the shear stress on a plane whose normal makes an angle of 120° with the x-axis. [Use set square, compass and protractor. Geometric construction lines carries marks. Graph paper will not be supplied]	5
4	(a)	In the (σ_1, σ_3) plane the failure criteria has the equation of a straight line $\sigma_1 = 5 + 2.5\sigma_3$. The induced stress in the vicinity of an excavation was (σ_1, σ_3) = (32, 10) MPa. Determine the factor of safety of the point mathematically using the concept of Mohr's Circle.	5

Q. No.	Question		Marks																											
4	(b)	The failure criteria of a rock is given by $\tau = 1.10 + \sigma_n \tan(40.12^\circ)$ MPa. If the state of stress in the rock is $\sigma_1 = 34.47$ MPa and $\sigma_3 = 8.96$ MPa, determine the value of the pore pressure that will result into fracturing of the rock.	5																											
5	(a)	In hydraulic fracturing experiment, the fracture initiates along the major principal stress direction. Prove the proposition mathematically using neat and labeled diagram.	5																											
	(b)	Determine the weighted average Q-value from the data given below. <table border="1" data-bbox="470 488 1187 943"> <thead> <tr> <th>Parameter of Q</th><th>Poorest value</th><th>Most typical value</th><th>Maximum value</th></tr> </thead> <tbody> <tr> <td>RQD</td><td>35</td><td>70</td><td>85</td></tr> <tr> <td>J_n</td><td>15</td><td>6</td><td>--</td></tr> <tr> <td>J_r</td><td>1.5</td><td>3</td><td>4</td></tr> <tr> <td>J_a</td><td>3</td><td>1</td><td>0.5</td></tr> <tr> <td>J_w</td><td>0.2</td><td>0.33</td><td>0.66</td></tr> <tr> <td>SRF</td><td>7.5</td><td>5</td><td>2.5</td></tr> </tbody> </table>	Parameter of Q	Poorest value	Most typical value	Maximum value	RQD	35	70	85	J_n	15	6	--	J_r	1.5	3	4	J_a	3	1	0.5	J_w	0.2	0.33	0.66	SRF	7.5	5	2.5
Parameter of Q	Poorest value	Most typical value	Maximum value																											
RQD	35	70	85																											
J_n	15	6	--																											
J_r	1.5	3	4																											
J_a	3	1	0.5																											
J_w	0.2	0.33	0.66																											
SRF	7.5	5	2.5																											

The relevant formula for solving the problem is given as under.
(The symbols have their usual meaning)

$$(\vec{p}) = [\tau] [\hat{n}],$$

$$\tau' = L\tau L^T$$

$$L = \begin{pmatrix} \cos \phi & \sin \phi \\ -\sin \phi & \cos \phi \end{pmatrix} \text{ for 2D}$$

$$L = \begin{bmatrix} \cos(\angle x'ox) & \cos(\angle x'oy) & \cos(\angle x'oz) \\ \cos(\angle y'ox) & \cos(\angle y'oy) & \cos(\angle y'oz) \\ \cos(\angle z'ox) & \cos(\angle z'oy) & \cos(\angle z'oz) \end{bmatrix} \text{ for 3D}$$

$$\sigma_\theta = \sigma_H + \sigma_h - p - 2(\sigma_H - \sigma_h)\cos 2\theta$$

Soln 1 The Vector be defined as

$$\hat{n} = N_x \hat{i} + N_y \hat{j} + N_z \hat{k}$$

as per the figure and the Co-ordinate Systems provided

$$N_x = \sin \delta \sin \theta$$

$$N_y = \sin \delta \cos \theta$$

$$N_z = \cos \delta$$

$$\text{Given } \delta = 30^\circ \quad \theta = 60^\circ$$

$$\therefore N_x = \sin 30^\circ \sin 60^\circ = \frac{\sqrt{3}}{4}$$

$$N_y = \sin 30^\circ \cos 60^\circ = \frac{1}{4}$$

$$N_z = \cos 30^\circ = \frac{\sqrt{3}}{2}$$

Check

$$\begin{aligned} & N_x^2 + N_y^2 + N_z^2 \\ &= \frac{3}{16} + \frac{1}{16} + \frac{3}{4} = 1 \quad [\text{Verified squares of} \\ & \quad \text{dc's is unity}] \end{aligned}$$

Now

$$\vec{P} = [T] \hat{n} = \begin{bmatrix} 5 & 2 & 3 \\ 2 & 12 & 4 \\ 3 & 4 & 7 \end{bmatrix} \begin{bmatrix} \sqrt{3}/4 \\ 1/4 \\ \sqrt{3}/2 \end{bmatrix}$$

$$= \begin{bmatrix} 5.26 \\ 6.89 \\ 8.36 \end{bmatrix}$$

$$\sigma = \vec{P} \cdot \hat{n} = \begin{bmatrix} \sqrt{3}/4 \\ 1/4 \\ \sqrt{3}/2 \end{bmatrix} \cdot \begin{bmatrix} 5.26 \\ 6.89 \\ 8.36 \end{bmatrix} = 11.24 \text{ MPa}$$

Soln(2) (a)

Normal Strain :

The normal strain in say x -direction in case of two dimensions is defined as the fractional shortening of an ^{infinitesimal} line that is initially oriented along the

$$x\text{-axis. } \epsilon_{xx} = \frac{\partial u}{\partial x} \quad || \quad \epsilon_{yy} = \frac{\partial v}{\partial y}$$

Shear Strain

The Shear Strain is defined as half the increase in angle formed by two infinitesimal line ^{segment} that are initially oriented along x & y axes:

$$\epsilon_{xy} = \frac{1}{2} \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right)$$

The Strain matrix in differential form is given as

$$\epsilon = \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{1}{2} \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right) \\ \frac{1}{2} \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) & \frac{\partial v}{\partial y} \end{bmatrix}$$

Soln-(2) (b)

Under the rotation of Co-ordinate system

we have $[X'] = [L][X]$

$$\text{or, } \left. \begin{aligned} x' &= x \cos \theta + y \sin \theta \\ y' &= -x \sin \theta + y \cos \theta \end{aligned} \right\} \text{--- (1)}$$

The inverse relation is,

$$[X] = [L]^T [X']$$

$$\text{or, } \left. \begin{aligned} x &= x' \cos \theta - y' \sin \theta \\ y &= x' \sin \theta + y' \cos \theta \end{aligned} \right\} \text{--- (2)}$$

Since $\vec{u}(x, y)$ is a vector therefore,

$$\vec{u}(x, y) = u(x, y) \hat{i} + v(x, y) \hat{j}$$

The relations (1) & (2) when applied to displacement vector becomes,

$$\left. \begin{aligned} u' &= u \cos \theta + v \sin \theta \\ v' &= -u \sin \theta + v \cos \theta \\ \text{or } u &= u' \cos \theta - v' \sin \theta \\ v &= u' \sin \theta + v' \cos \theta \end{aligned} \right\} \text{--- (3)}$$

Now consider the component of strain

eg: $\epsilon_{x'x'}$ & by chain rule we have,

$$\left. \begin{aligned} \epsilon_{x'x'} = \frac{\partial u'}{\partial x'} &= \frac{\partial u'}{\partial u} \cdot \frac{\partial u}{\partial x} \frac{\partial x}{\partial x'} + \frac{\partial u'}{\partial u} \cdot \frac{\partial u}{\partial y} \frac{\partial y}{\partial x'} \\ &+ \frac{\partial u'}{\partial v} \cdot \frac{\partial v}{\partial x} \frac{\partial x}{\partial x'} + \frac{\partial u'}{\partial v} \cdot \frac{\partial v}{\partial y} \frac{\partial y}{\partial x'} \end{aligned} \right\} \text{--- (4)}$$

Now comparing the partial derivatives of Eqn ①, ② & ③ and substituting them into Eqn ④ we have

$$\begin{aligned} \epsilon_{x'x'} &= \frac{\partial u'}{\partial x'} = \cos\theta \frac{\partial u}{\partial x} \cos\theta + \cos\theta \frac{\partial u}{\partial y} \sin\theta \\ &\quad + \sin\theta \frac{\partial v}{\partial x} \cos\theta + \sin\theta \frac{\partial v}{\partial y} \sin\theta \\ &= \frac{\partial u}{\partial x} \cos^2\theta + \frac{\partial u}{\partial y} \sin\theta \cos\theta + \frac{\partial v}{\partial x} \sin\theta \cos\theta \\ &\quad + \frac{\partial v}{\partial y} \sin^2\theta \end{aligned}$$

$$\text{or, } \epsilon_{x'x'} = \epsilon_{xx} \cos^2\theta + 2\epsilon_{xy} \sin\theta \cos\theta + \epsilon_{yy} \sin^2\theta.$$

$\downarrow$ ⑤

We observe that Eqn ⑤ is similar to Stress transformation law as in 2D for T_{xx}

Similar analysis of other components will reveal that

$$\boxed{\epsilon' = L \epsilon L^T} \quad \text{--- ⑥}$$

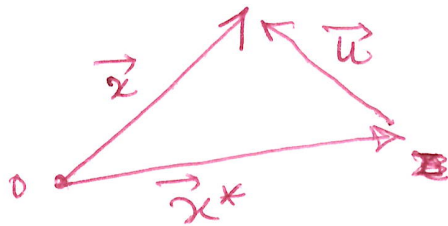
The appearance of 'L' matrix twice in the transformation equation (6) proves that

The Strain is a ~~1st~~ second order tensor.

Soln. 2 (c)

In terms of rock mechanics

Convention the displacement vector is interpreted as the vector that points from New position towards the original position.



$$\therefore \vec{x}^* = \vec{x} - \vec{u}$$

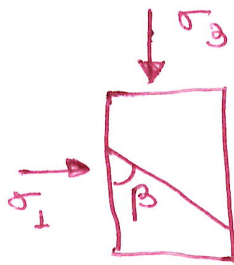
$$\text{we have } \vec{x} = (2, 5, -4)$$

$$\begin{aligned} \vec{u}_{(2,5,-4)} &= [9 \times 2^3 + 4] \hat{i} + [15 \times 5 + 3 \times 2^2] \hat{j} \\ &\quad + [6 \times (5)^2 + 2(-4)^2] \hat{k} \\ &= (76 \hat{i} + 87 \hat{j} + 182 \hat{k}) \end{aligned}$$

$$\begin{aligned} \therefore \vec{x}^* &= (2, 5, -4) - (76, 87, 182) \\ &= (-74, -82, -186) \end{aligned}$$

= deformed position at the given point

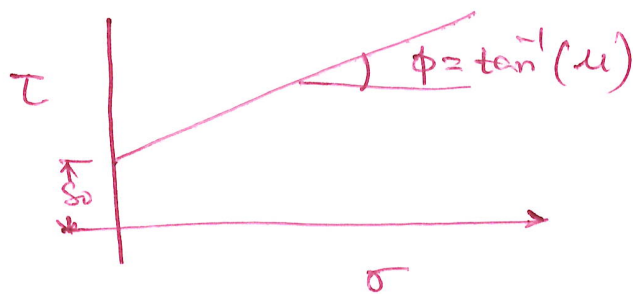
Soln- 3(a)



When a rock specimen is subjected to biaxial loading; and if the stress levels are such that ~~it~~ they are sufficient for fracturing the rock, then the failure plane makes an angle ' β ' with the σ_3 axis.

The failure along the plane is governed by the Coulomb's Criteria

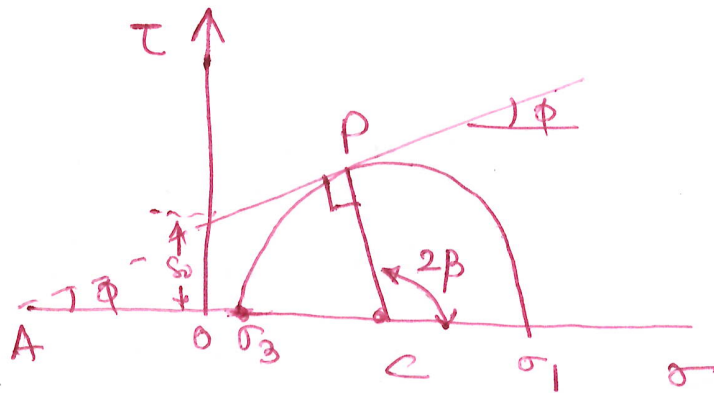
$$\tau = s_0 + \mu \sigma \quad \text{--- (1)}$$



At the failure, the Mohr's Circle of (σ_1 & σ_3) touches the failure criterion line tangentially.

In other words, the normal stress and shear stress on the plane at the time of failure satisfy the Equation (1) given above.

The Mohr's Circle for the above mentioned situation can be drawn as shown in next page.



The relationship between β & ϕ can be extracted from the above figure

$$2\beta = \frac{\pi}{2} + \phi$$

$$\text{or, } \boxed{\beta = \frac{\pi}{4} + \frac{\phi}{2}}$$

Now from $\triangle ACP$.

$$\begin{aligned} CP &= AC \sin \phi \\ &= (AO + OC) \sin \phi = AO \sin \phi + OC \sin \phi \\ &= S_0 \cot \phi \sin \phi + OC \sin \phi \\ &= S_0 \cos \phi + OC \sin \phi \end{aligned}$$

$$\text{but } CP = \frac{\sigma_1 - \sigma_3}{2} \quad \& \quad OC = \frac{\sigma_1 + \sigma_3}{2}$$

$$\Rightarrow \frac{\sigma_1 - \sigma_3}{2} = S_0 \cos \phi + \frac{\sigma_1 + \sigma_3}{2} \sin \phi$$

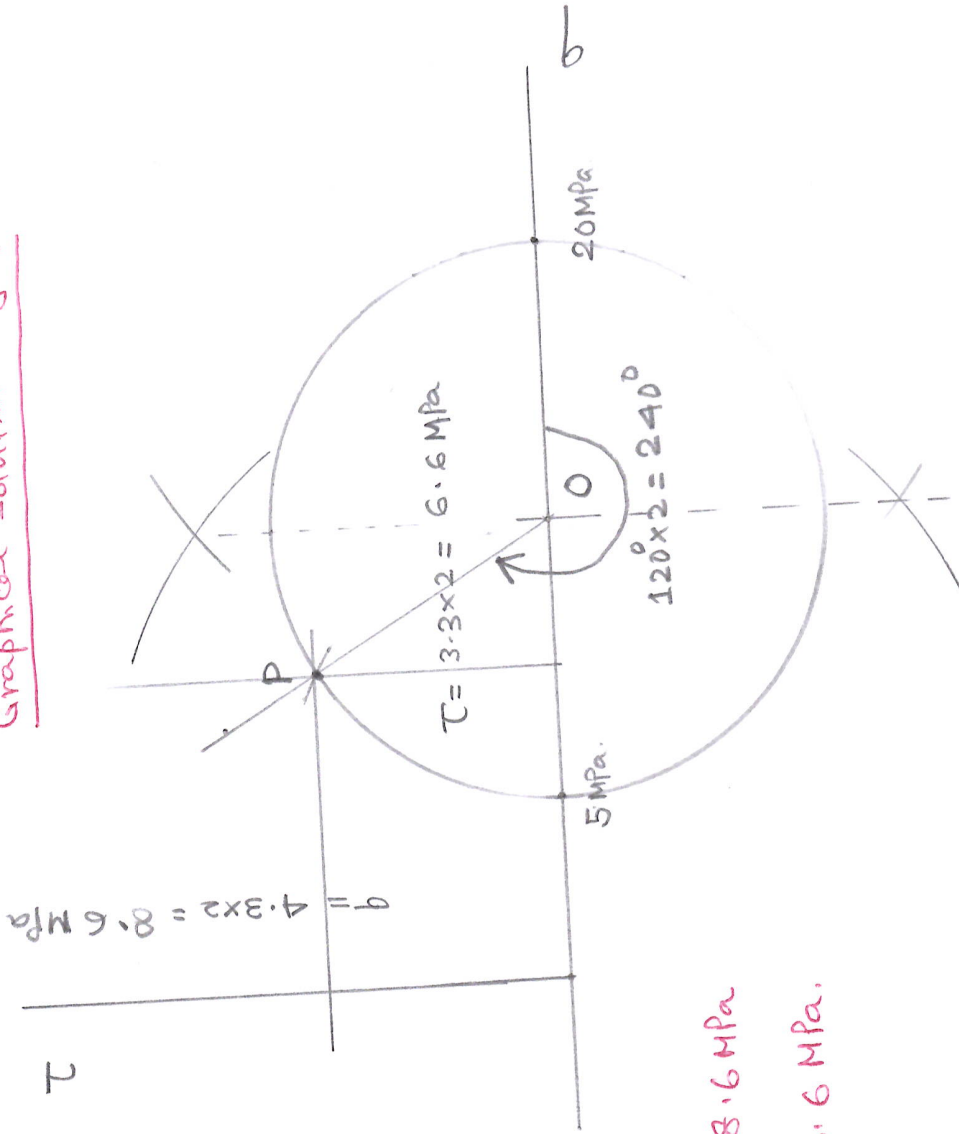
Rearranging the σ_1 & σ_3 term.

$$\frac{\sigma_1}{2} (1 - \sin \phi) = S_0 \cos \phi + \frac{\sigma_3}{2} (1 + \sin \phi)$$

$$\text{or, } \boxed{\sigma_1 = \frac{2 S_0 \cos \phi}{1 - \sin \phi} + \sigma_3 \left(\frac{1 + \sin \phi}{1 - \sin \phi} \right)}$$

Solution 3(b)

Graphical Solution Sought



Analytical Verification

$$\theta = 120^\circ$$

$$L = \begin{bmatrix} -0.5 & 0.866 \\ -0.866 & -0.5 \end{bmatrix}$$

$$T = \begin{bmatrix} 20 & 0 \\ 0 & 5 \end{bmatrix}$$

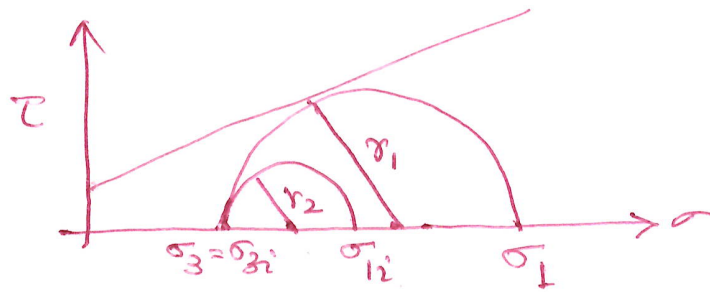
$$LTL^T = \begin{bmatrix} 8.75 & 6.5 \\ 6.5 & 16.25 \end{bmatrix}$$

$$\sigma \approx 8.75 \xrightarrow[\text{to}]{\text{close}} 8.6 \text{ MPa}$$

$$\tau \approx 6.5 \xrightarrow[\text{to}]{\text{close}} 6.6 \text{ MPa}$$

Verified.

Soln 4(a)



As per defn. of FOS using Mohr's Circle Concept

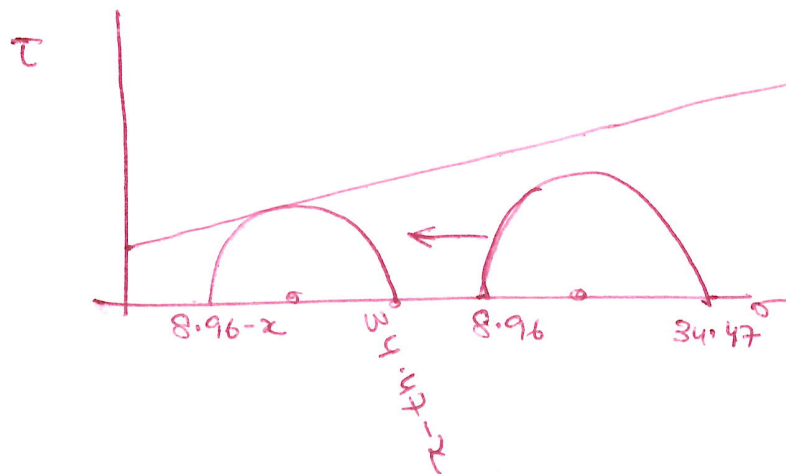
$$FOS = \frac{r_1}{r_2} = \left[\frac{\sigma_1 - \sigma_{3i}}{\sigma_{1i} - \sigma_{3i}} \right] \quad \text{At } \sigma_3 = \sigma_{3i}$$

$$\text{Given } (\sigma_{1i}, \sigma_{3i}) = (30, 10)$$

$$\begin{aligned} \sigma_1 &= 5 + 2.5 \sigma_{3i} \\ &= 5 + 2.5 \times 10 = 30 \end{aligned}$$

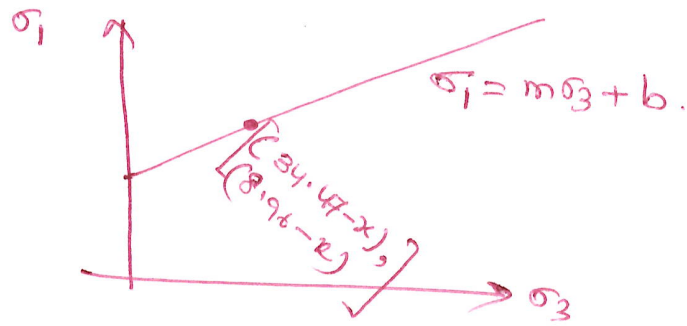
$$\therefore FOS = \frac{30 - 10}{32 - 10} = \frac{20}{22} = 0.9090$$

Soln. 4(b)



Let x = pore pressure that is going to result into fracturing the rock.

At this stage the Mohr's Circle will touch the failure line.



The pore-pressure (x) will lead to ^{reduction in} effective ~~reduction~~

Stress

$\therefore$ at the time of failure the effective stress will be.

$$(\sigma_{12} - x, \sigma_{31} - x) = (34.47 - x, 8.96 - x)$$

This effective stress will lie on the failure line at the time of failure initiation.

So Let's find m & b .

$$m = \frac{1 + \sin \phi}{1 - \sin \phi} = \frac{1 + \sin 40.12}{1 - \sin 40.12} = 4.62$$

$$b = \frac{2c \cos \phi}{1 - \sin \phi} = \frac{2 \times 1.1 \times \cos 40.12}{1 - \sin(40.12)} = 4.73$$

$$\sigma_1 = 4.62 \sigma_3 + 4.73$$

$$\text{or, } (34.47 - x) = 4.62(8.96 - x) + 4.73$$

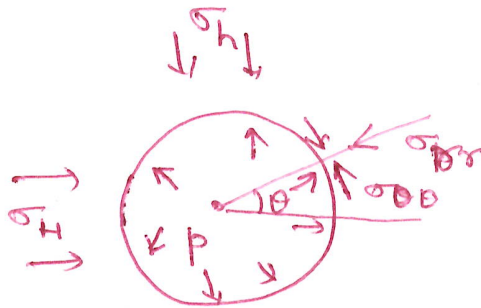
$$34.47 - x = 4.62 \times 8.96 - 4.62x + 4.73$$

$$3.62x = 11.65$$

$$\text{or, } x = \frac{11.65}{3.62} = 3.218$$

$$\approx 3.22 \text{ MPa.}$$

Soln. 5(a)

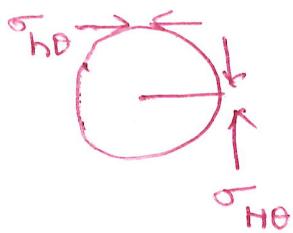


$$\sigma_r = p$$

$$\sigma_\theta = \sigma_H + \sigma_h - p - 2(\sigma_H - \sigma_h) \cos 2\theta$$

[Formula Given]

$$\forall \theta = 0^\circ \quad \sigma_{H\theta} = (3\sigma_h - \sigma_H) - p \quad \text{--- (1)}$$



$$\forall \theta = 90^\circ$$

$$\sigma_{h\theta} = (3\sigma_H - \sigma_h) - p \quad \text{--- (2)}$$

Now by Defn

$$\sigma_h < \sigma_H$$

$$\therefore 4\sigma_h < 4\sigma_H$$

$$\therefore (3\sigma_h - \sigma_H) < (3\sigma_H - \sigma_h)$$

$$\therefore (3\sigma_h - \sigma_H) - p < (3\sigma_H - \sigma_h) - p$$

$$\therefore \sigma_{H\theta} < \sigma_{h\theta}$$

$\Rightarrow \sigma_{H\theta}$ has less magnitude.

As long as $\sigma_{H\theta} > 0$ it is compressional. Now as the fluid pressure $\uparrow$ $\sigma_{H\theta} < 0$ (Tensile)

$$\therefore 3\sigma_h - \sigma_H - p < 0 \text{ [Tensile at high } p']$$

The breakage will occur at high ' p ' when the $\sigma_{H\theta}$ overcomes the rock tensile strength.

$\therefore$ The fracture will occur in the direction of σ_H .

Soln - 5(b)

Parameter Q	Poorest value (10%) Wt	Most Typical value (60%)	max Value (30%)	Wt. Avg
ROD.	35	70	85	71
J_n .	15	0.6	—	7.29
J_r .	1.5	3	4	3.15
J_a	3	1	0.5	1.05
J_w	0.2	0.33	0.66	0.416
SRF	7.5	5	2.5	4.5

$$Q = \frac{ROD}{J_n} \times \frac{J_r}{J_a} \times \frac{J_w}{SRF}$$

$$= \frac{71}{7.29} \times \frac{3.15}{1.05} \times \frac{0.416}{4.5}$$

$$= 2.701$$

$$\text{Wt. Avg of } J_n = \frac{(15 \times 0.1 + 6 \times 0.6)}{0.7}$$

$$= 7.29$$

ROD is 71 only ~~not~~ not to be rounded to next 5. [See: Barton's Paper].